Week 1 | Solve With Instructors

Q1) You are given a dataset $\mathbf{X}_{d\times n}$ with the following properties:

$$\text{Covariance Matrix: } C = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

$$\text{Eigenvectors of } C: \mathbf{w}_1 = \begin{bmatrix} -0.894 \\ 0.447 \end{bmatrix}, \mathbf{w}_2 = \begin{bmatrix} 0.447 \\ 0.894 \end{bmatrix}$$

Note to learners that the eigenvectors are not necessary to be in order with respect to their eigenvalues. That is, it is not necessary that $\mathbf{w}_1$ is first principal component. Answer the following questions based on the above data:

i) Which eigenvector, $\mathbf{w}_1$ or $\mathbf{w}_2$, represents the first principal component of the dataset?

Solution:

i) First principal component is the eigenvector having largest eigenvalue. To find the eigenvalue corresponds to a eigenvector, we will use the below formula

$$\lambda_i = \mathbf{w}_i^T C \mathbf{w}_i$$

$$\begin{aligned}\lambda_1 &= \begin{bmatrix} -0.894 & 0.447 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} -0.894 \\ 0.447 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 \end{bmatrix} \begin{bmatrix} -0.894 \\ 0.447 \end{bmatrix} \\ &= 0\end{aligned}$$

$$\begin{aligned}\lambda_2 &= \begin{bmatrix} 0.447 & 0.894 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 0.447 \\ 0.894 \end{bmatrix} \\ &= \begin{bmatrix} 2.235 & 4.47 \end{bmatrix} \begin{bmatrix} 0.447 \\ 0.894 \end{bmatrix} \\ &\approx 5\end{aligned}$$

As $\lambda_2 > \lambda_1$, $\mathbf{w}_2$ is the first principal component.

Reference data from previous question:

$$\text{Covariance Matrix: } C = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

$$\text{Eigenvectors of } C: \mathbf{w}_1 = \begin{bmatrix} -0.894 \\ 0.447 \end{bmatrix}, \mathbf{w}_2 = \begin{bmatrix} 0.447 \\ 0.894 \end{bmatrix}$$

ii) How much of variance in the data is explained by the first principal component?

Solution:

ii) The largest eigenvalue corresponds to the variance captured across first principal component. We got this before as 5.

Reference data from previous question:

$$\text{Covariance Matrix: } C = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

$$\text{Eigenvectors of } C: \mathbf{w}_1 = \begin{bmatrix} -0.894 \\ 0.447 \end{bmatrix}, \mathbf{w}_2 = \begin{bmatrix} 0.447 \\ 0.894 \end{bmatrix}$$

iii) For the given dataset $\mathbf{X}_{d\times n}$, the PCA-transformed dataset (by only taking the scalar projection) is $\mathbf{Y}_{d\times n}$. What is the covariance matrix for the PCA-transformed data? (**Hint:** Eigenvalue decomposition)

Solution:

iii) For a data point $\mathbf{x}_i$, the scalar projection is given by

$$\widetilde{\mathbf{x}_i} = \begin{bmatrix} \mathbf{w}_1^T \mathbf{x}_i \\ \mathbf{w}_2^T \mathbf{x}_i \end{bmatrix}$$

Note that $\forall \ \|\mathbf{w}_i\| = 1$. Here $\mathbf{w}_1$ refers to the first principal component not the one given in the question.

Consider a matrix $\mathbf{W}$ consisting eigenvectors as columns, arranged in the descending order based on their eigenvalues.

$$\mathbf{W} = \begin{bmatrix} | & | \\ \mathbf{w}_1 & \mathbf{w}_2 \\ | & | \end{bmatrix} = \begin{bmatrix} 0.447 & -0.894 \\ 0.894 & 0.447 \end{bmatrix}$$

We can write $\widetilde{\mathbf{x}_i}$ as

$$\widetilde{\mathbf{x}_i} = \begin{bmatrix} \mathbf{w}_1^T \mathbf{x}_i \\ \mathbf{w}_2^T \mathbf{x}_i \end{bmatrix} = \begin{bmatrix} - & \mathbf{w}_1^T & - \\ - & \mathbf{w}_2^T & - \end{bmatrix} \begin{bmatrix} | \\ \mathbf{x}_i \\ | \end{bmatrix} = \mathbf{W}^T \mathbf{x}_i$$

Let us now construct the matrix $\mathbf{Y}_{d\times n}$

$$\mathbf{Y}_{d\times n} = \begin{bmatrix} | & & | \\ \widetilde{\mathbf{x}_1} & \cdots & \widetilde{\mathbf{x}_n} \\ | & & | \end{bmatrix}$$
$$= \begin{bmatrix} \mathbf{w}_1^T\mathbf{x}_1 & \cdots & \mathbf{w}_1^T\mathbf{x}_n \\ \mathbf{w}_2^T\mathbf{x}_1 & \cdots & \mathbf{w}_2^T\mathbf{x}_n \end{bmatrix}$$

Using the result, $\widetilde{\mathbf{x}_i} = \begin{bmatrix} \mathbf{w}_1^T\mathbf{x}_i \\ \mathbf{w}_2^T\mathbf{x}_i \end{bmatrix} = \begin{bmatrix} - & \mathbf{w}_1^T & - \\ - & \mathbf{w}_2^T & - \end{bmatrix} \begin{bmatrix} | \\ \mathbf{x}_i \\ | \end{bmatrix} = \mathbf{W}^T\mathbf{x}_i$, we can write $\mathbf{Y}$ as

$$\mathbf{Y} = \begin{bmatrix} \mathbf{w}_1^T\mathbf{x}_1 & \cdots & \mathbf{w}_1^T\mathbf{x}_n \\ \mathbf{w}_2^T\mathbf{x}_1 & \cdots & \mathbf{w}_2^T\mathbf{x}_n \end{bmatrix} = \mathbf{W}^T\mathbf{X}$$

So, our transformed data matrix $\mathbf{Y}$ is $\mathbf{W}^T\mathbf{X}$. Let us find the covariance matrix for this data matrix

$$
\begin{aligned}
\widetilde{C} &= \frac{1}{n}\mathbf{Y}\mathbf{Y}^T \\
&= \frac{1}{n}\left(\mathbf{W}^T\mathbf{X}\right)\left(\mathbf{W}^T\mathbf{X}\right)^T \\
&= \frac{1}{n}\mathbf{W}^T\mathbf{X}\mathbf{X}^T\mathbf{W} \\
&= \mathbf{W}^T\left(\frac{1}{n}\mathbf{X}\mathbf{X}^T\right)\mathbf{W} \\
&= \mathbf{W}^T C\mathbf{W}
\end{aligned}
$$

From eigenvalue decomposition, we know $\mathbf{W}^T C\mathbf{W}$ is equal to a diagonal matrix where diagonal entries are the eigenvalues of the matrix C

$$
\mathbf{W}^T C\mathbf{W} = \begin{bmatrix} 5 & 0 \\ 0 & 0 \end{bmatrix}
$$

Q2) Which of the following statements are true with respect to PCA?

- [ ] All principal components are uncorrelated.
- [ ] PCA is primarily used for the task of dimensionality reduction.
- [ ] The covariance matrix used in PCA is always symmetric.
- [ ] The eigenvector with smallest positive eigenvalue is the first principal component.

Solution:
Let's evaluate each statement regarding Principal Component Analysis (PCA):

1. All principal components are uncorrelated.
   True. One of the key properties of PCA is that the principal components are orthogonal to each other, which means they are uncorrelated.

2. PCA is primarily used for the task of dimensionality reduction.
   True. PCA is commonly used for dimensionality reduction by transforming the original variables into a smaller number of uncorrelated variables (the principal components) while retaining most of the variance.

3. The covariance matrix used in PCA is always symmetric.
   True. The covariance matrix is defined as $\frac{1}{n}\mathbf{X}\mathbf{X}^T$ which is always symmetric. Another way to look at it is $Cov(x, y) = Cov(y, x)$.

4. The eigenvector with the smallest positive eigenvalue is the first principal component.
   False. The first principal component corresponds to the eigenvector with the largest eigenvalue, as it explains the most variance.

Q3) Standard PCA was applied on a centered dataset in $\mathbb{R}^6$. All the eigenvalues of the covariance matrix of this centered dataset are given below in decreasing order with $\lambda > 1$.

$$\lambda^5, \lambda^4, \lambda^3, \lambda^2, \lambda, 1$$

If the sum of the variances along the first three principal components is exactly 88.88% of the total variance, what is the value of λ? Enter the nearest integer as your answer.

Solution:

Given

$$\frac{\lambda^5+\lambda^4+\lambda^3}{\lambda^5+\lambda^4+\lambda^3+\lambda^2+\lambda+1}=0.8888$$

Let denote $\lambda^5+\lambda^4+\lambda^3=a$ and $\lambda^2+\lambda+1=b$

$$\frac{a}{a+b}=0.8888$$

$$a=0.8888(a+b)$$

$$a(1-0.888)=0.8888b$$

$$0.1112a=0.8888b$$

$$a=7.99b$$

Now, replace a with $\lambda^5+\lambda^4+\lambda^3$ and b with $\lambda^2+\lambda+1$

$$\lambda^5 + \lambda^4 + \lambda^3 = 7.99\left(\lambda^2 + \lambda + 1\right)$$

$$\lambda^3\left(\lambda^2 + \lambda + 1\right) = 7.99\left(\lambda^2 + \lambda + 1\right)$$

Notice that $\lambda^2 + \lambda + 1$ will always be greater than 0 as λ is the eigen value of a covariance matrix which is a positive semi-definite matrix so it's eigen values will be greater than or equal to 0.

$$\lambda^3\cancel{\left(\lambda^2 + \lambda + 1\right)} = 7.99\cancel{\left(\lambda^2 + \lambda + 1\right)}$$

$$\lambda^3 = 7.99$$
$$\lambda \approx 2$$

Q4) Consider a dataset of 100 points where the i_{th} data point is given by:

$$\mathbf{x}_i = a_i \cdot \begin{bmatrix} 3 \\ 1 \\ 0 \\ 2 \end{bmatrix} + b_i \cdot \begin{bmatrix} 1 \\ 0 \\ 3 \\ 0 \end{bmatrix}$$

where a_i and b_i are real numbers such that $\sum_{i=0}^{100} a_i = \sum_{i=0}^{100} b_i = 0$. Standard PCA is performed on this dataset. If the top two principal components are retained and used to reconstruct the dataset, what is the reconstruction error?

Hint: Think about what happens in $\mathbb{R}^2$ or $\mathbb{R}^3$ for similar situation and extend this idea to $\mathbb{R}^4$.

Solution:
To solve this problem, we need to understand the dataset structure and how PCA works for reconstruction.

Dataset Structure:

Each data point $\mathbf{x}_i$ is a linear combination of two fixed vectors:

$$\mathbf{x}_i = a_i \cdot \begin{bmatrix} 3 \\ 1 \\ 0 \\ 2 \end{bmatrix} + b_i \cdot \begin{bmatrix} 1 \\ 0 \\ 3 \\ 0 \end{bmatrix}$$

This means all the data points $\mathbf{x}_i$ lie in the 2-dimensional subspace spanned by these two vectors:

$$v_1 = \begin{bmatrix} 3 \\ 1 \\ 0 \\ 2 \end{bmatrix}, \; v_2 = \begin{bmatrix} 1 \\ 0 \\ 3 \\ 0 \end{bmatrix}$$

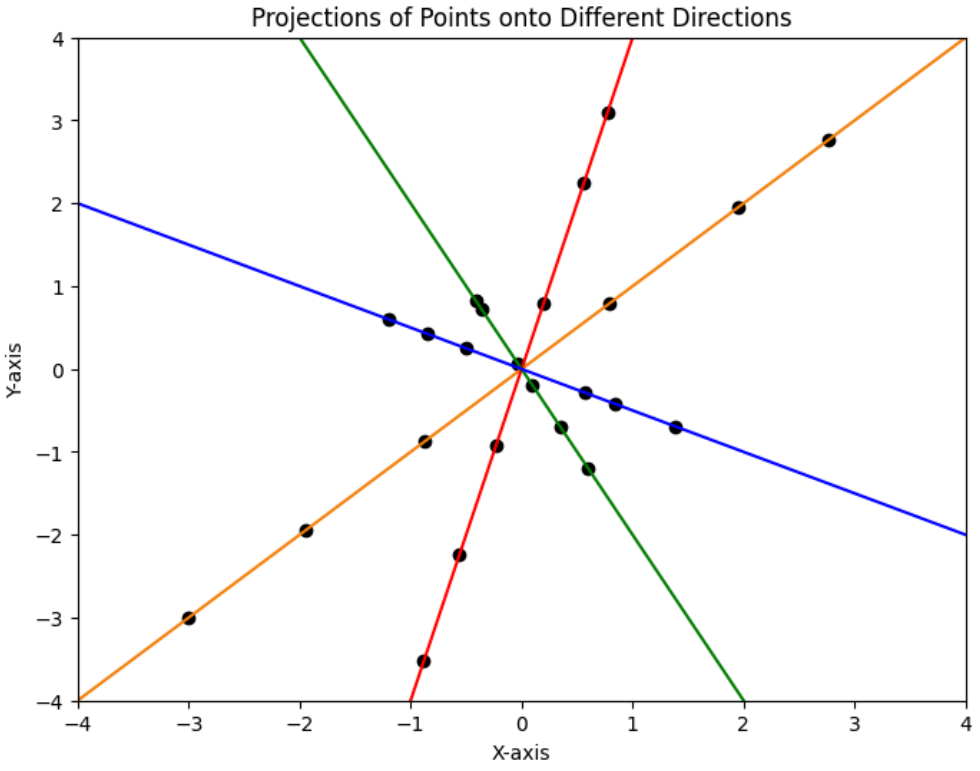
The key insight here is that the dataset already lies in a 2-dimensional subspace of $\mathbb{R}^4$ meaning that any point in the dataset can be represented by a combination of v_1 and v_2.

PCA and Reconstruction:

PCA aims to find the principal components (directions) that capture the most variance in the data. In this case, the top two principal components will align with the vectors v_1 and v_2 since they already define the subspace containing all the data points.

When the top two principal components are used for reconstruction, the data points will be perfectly reconstructed, because the original data lies entirely within the subspace defined by these two components. Hence, no information is lost during this reconstruction.

Q5) The dataset contains 6 points in $\mathbb{R}^2$. The black dotted points are the projection of these data points onto different lines. Assume none of the points are in any of the four lines given below. Which of the following lines could possibly be the $\mathbf{w_1}$, where w_1 represents the first principal component?



Solution:

The projections onto the first principal component shouldn't crowd up. From the picture, we can see that projections onto the orange line are least crowded.

Q6) There are 100 data points in $\mathbb{R}^{10}$. λ_i is the ith largest eigenvalue. The following observations are made:

- $\sum_{i=1}^{10} \lambda_i = 200$
- $\sum_{i=1}^{4} \lambda_i = 140$
- $\sum_{i=1}^{6} \lambda_i = 180$

We wish to retain 90% percent of the information in the dataset. How many numbers do we need to store?

Solution:

- To retain 90% of the information in the dataset, we will have to store the first 6 principal components.
- Every principal component is a vector in $\mathbb{R}^{10}$. For 6 principal components, we would have to store $6 \times 10 = 60$ numbers.
- For every data point x_i, we would have to store $x_i^T w_1, x_i^T w_2, \ldots, x_i^T w_6$, where $w_1, w_2, \ldots, w_6$ are the first 6 principal components.
- For 100 data points, we will have to store $6 \times 100 = 600$ numbers.
- Overall, we have to store 660 numbers.

Q7) There are six data points $\{x_1, x_2, ..., x_6\}$. The length of the projections of these six data points onto $\mathbf{w_1}$ are $\{8, 7, 3, 5, 2, 4\}$ respectively. Here, w_1 is the first principal component. Find the variance along the first principal component.

Solution:

For a data point x_i, the projection onto the first principal component w_1 is $\left(x_i^T w_1\right) w_1$.

Length of $\left(x_i^T w_1\right) w_1$ is given by,

$$||\left(x_i^T w_1\right) w_1|| = |x_i^T w_1| \times ||w_1|| = |x_i^T w_1|$$

The variance along the first principal component is $\frac{1}{6} \sum_{i=1}^{6} \left(x_i^T w_i\right)^2$

From the given information we can see that the following are true.

a) $|x_1^T w_1| = 8$. Therefore, $\left(x_1^T w_1\right)^2 = 64$

b) $|x_2^T w_1| = 7$. Therefore, $\left(x_2^T w_1\right)^2 = 49$

c) $|x_3^T w_1| = 3$. Therefore, $\left(x_3^T w_1\right)^2 = 9$

d) $|x_4^T w_1| = 5$. Therefore, $\left(x_4^T w_1\right)^2 = 25$

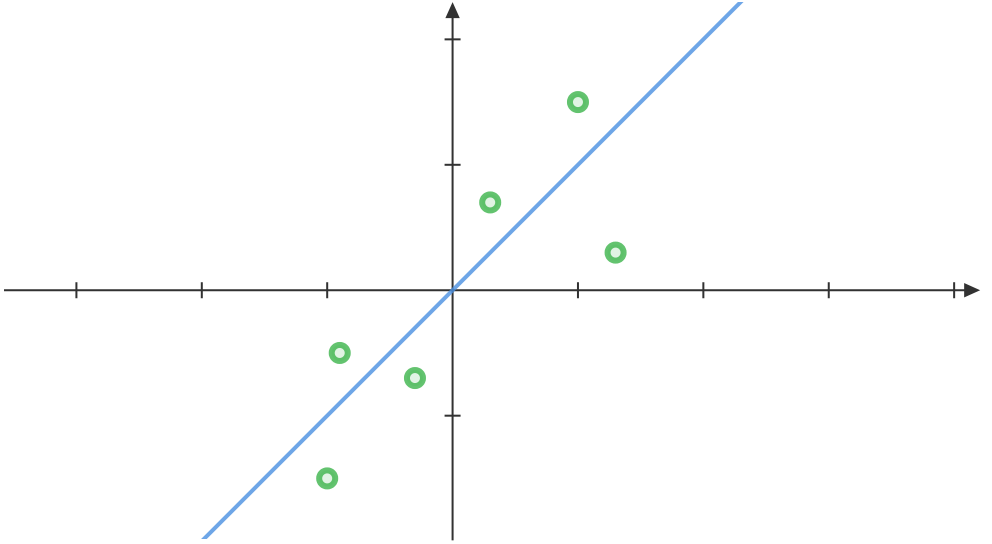
e) $|x_5^T w_1| = 2$. Therefore, $\left(x_5^T w_1\right)^2 = 4$

f) $|x_6^T w_1| = 4$. Therefore, $\left(x_6^T w_1\right)^2 = 16$

Therefore,

$$\frac{1}{6}\sum_{i=1}^{6}\left(x_i^T w_i\right)^2 = \frac{1}{6}(64+49+9+25+4+16) = \frac{1}{6}(167) = 27.83$$

Q8) There are 6 data points $\{x_1, x_2, ..., x_6\}$ in $\mathbb{R}^2$. The length of the projections of these six data points onto $\mathbf{w_2}$ are $\{1, 2, 1, 3, 2, k\}$ respectively. Here, w_2 is the second principal component. The average error due to projection onto $\mathbf{w_1}$ is $\frac{28}{6}$. Here, w_1 is the first principal component is. Find k.



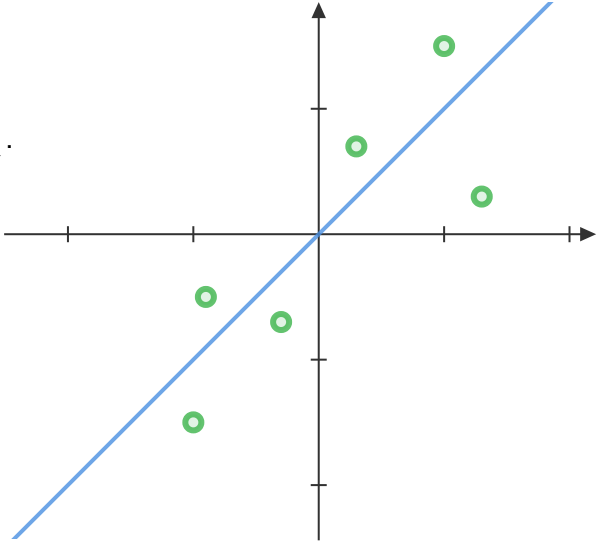
Solution:

Our data points are elements of $\mathbb{R}^2$. Let the blue line be $\mathbf{w_1}$

Here black lines represent the errors of each data point.
For a data point x_i, its error vector after projection onto $\mathbf{w_1}$ is given by $x_i - \left(x_i^T w_1\right) w_1$.

$x_i - \left(x_i^T w_1\right) w_1 \rightarrow$ This is the vector used for finding $\mathbf{w_2}$.

Therefore, the length of these error vectors are the length of the projections onto $\mathbf{w_2}$.



Therefore,

a) $\left|\left|x_1 - \left(x_1^T w_1\right) w_1\right|\right| = 1$

b) $\left|\left|x_2 - \left(x_2^T w_1\right) w_1\right|\right| = 2$

c) $\left|\left|x_3 - \left(x_3^T w_1\right) w_1\right|\right| = 1$

d) $\left|\left|x_4 - \left(x_4^T w_1\right) w_1\right|\right| = 3$

e) $\left|\left|x_5 - \left(x_5^T w_1\right) w_1\right|\right| = 2$

f) $\left|\left|x_6 - \left(x_6^T w_1\right) w_1\right|\right| = k$

The loss due to the projection onto $\mathbf{w_1}$ is given as $^{28}/_{6}$.

This is given by :

$$\begin{aligned}
\frac{1}{6}\sum_{i=1}^{6}\left|\left|x_1 - \left(x_1^T w_1\right) w_1\right|\right|^2 &= \frac{28}{6} \\
\frac{1}{6}\left(1^2 + 2^2 + 1^2 + 3^2 + 2^2 + k^2\right) &= \frac{28}{6} \\
1 + 4 + 1 + 9 + 4 + k^2 &= 28 \\
19 + k^2 &= 28 \\
k^2 &= 9 \\
k &= 3
\end{aligned}$$

k refers to the length, therefore, always greater than or equal to 0.

Q9) Consider a dataset that has 4 points in $\mathbb{R}^2$ that lie on a line passing through the origin:

$$D = \{(-2,-6),\ (-1,-3),\ (1,3),\ (2,6)\}$$

Based on the above data, answer the given sub-questions.

i) Find a representative w for this dataset such that $\|w\| = 1$.

Solution:

We have four points in 2D space that lie on a line passing through the origin:

$$D = \{(-2,-6),\ (-1,-3),\ (1,3),\ (2,6)\}$$

These points lie on the line $y = 3x$, so we expect that performing Principal Component Analysis (PCA) will capture the direction of this line.

Part (a): Find a unit vector w that represents the dataset

Step 1: The points lie along the line $y = 3x$. We can take any point in D as representative and normalize it since the norm of the representative should be one:

$$v = (2,6)$$

Step 2: Now, we need to normalize this vector (make its length 1). The length of $v = (2, 6)$ is:

$$\|v\| = \sqrt{2^2 + 6^2} = \sqrt{40}$$

Step 3: We normalize v by dividing each component by its length:

$$w = \left(\frac{1}{\sqrt{10}}, \frac{3}{\sqrt{10}} \right)$$

This is the representative for the dataset D.

ii) If standard PCA is performed on this dataset, what is the variance along the first principal component?

Solution:

Part (b): What is the variance along the first principal component?

The first principal component is along the direction of w. To find the variance, we need to project each point in D onto w and compute the variance of the projections.

Step 1: We project each point onto w using the formula for dot products. The projections are:

$$(-2, -6) \cdot w = -2\sqrt{10}$$

$$(-1, -3) \cdot w = -\sqrt{10}$$

$$(1, 3) \cdot w = \sqrt{10}$$

$$(2, 6) \cdot w = 2\sqrt{10}$$

Step 2: The mean of these projections is 0. The variance is calculated by averaging the squared distances of each projection from the mean:

$$\text{Variance} = \frac{(-2\sqrt{10})^2 + (-\sqrt{10})^2 + (\sqrt{10})^2 + (2\sqrt{10})^2}{4} = 25$$

Thus, the variance along the first principal component is 25.

iii) If standard PCA is performed on this dataset, what is the variance along the second principal component?

Solution:

Part (c): What is the variance along the second principal component?

Since all the points lie exactly on the line $y = 3x$, there is no spread in any direction perpendicular to this line. Therefore, the variance along the second principal component is 0.

- (a): The unit vector w is $\left(\frac{1}{\sqrt{10}}, \frac{3}{\sqrt{10}}\right)$.
- (b): The variance along the first principal component is 25.
- (c): The variance along the second principal component is 0.

Q10) Consider the following data points x_1, x_2, x_3 to answer the following questions:

$$
\begin{aligned}
x_1 &= (2, 2) \\
x_2 &= (1, 2) \\
x_3 &= (2, 1)
\end{aligned}
$$

i) What is the mean of the given data points x_1, x_2, x_3?

Solution:

Given Data Points:

$$x_1 = (2, 2),\ x_2 = (1, 2),\ x_3 = (2, 1)$$

i) **Mean of the data points**

The mean vector $\overline{x}$ is calculated as:

$$\overline{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

where $n = 3$.

So,

$$\overline{x} = \frac{1}{3}(x_1 + x_2 + x_3) = \frac{1}{3}((2,2) + (1,2) + (2,1))$$

$$\overline{x} = \frac{1}{3}((2+1+2, 2+2+1)) = \frac{1}{3}(5,5)$$

$$\overline{x} = \left(\frac{5}{3}, \frac{5}{3}\right)$$

Thus, the mean vector is:

$$\overline{x} = \left(\frac{5}{3}, \frac{5}{3}\right)$$

ii) Find the covariance matrix C

$$\text{where} \quad C = \frac{1}{n}\sum_{i=1}^{n}(x-\overline{x})(x-\overline{x})^T$$

Solution:

ii) Covariance matrix C

The covariance matrix is given by:

$$C = \frac{1}{n}\sum_{i=1}^{n}(x_i - \overline{x})(x_i - \overline{x})^T$$

First, calculate $x_i - \overline{x}$ for each i:

$$x_1 - \overline{x} = (2,2) - \left(\frac{5}{3}, \frac{5}{3}\right) = \left(\frac{1}{3}, \frac{1}{3}\right)$$

$$x_2 - \overline{x} = (1,2) - \left(\frac{5}{3}, \frac{5}{3}\right) = \left(-\frac{2}{3}, \frac{1}{3}\right)$$

$$x_3 - \overline{x} = (2,1) - \left(\frac{5}{3}, \frac{5}{3}\right) = \left(\frac{1}{3}, -\frac{2}{3}\right)$$

Next, calculate $(x_i - \overline{x})(x_i - \overline{x})^T$ for each data point:

$$(x_1 - \overline{x})(x_1 - \overline{x})^T = \left(\frac{1}{3}, \frac{1}{3}\right)\left(\frac{1}{3}, \frac{1}{3}\right)^T = \left(\frac{1}{9}, \frac{1}{9}, \frac{1}{9}, \frac{1}{9}\right)$$

$$(x_2 - \overline{x})(x_2 - \overline{x})^T = \left(-\frac{2}{3}, \frac{1}{3}\right)\left(-\frac{2}{3}, \frac{1}{3}\right)^T = \left(\frac{4}{9}, -\frac{2}{9}, -\frac{2}{9}, \frac{1}{9}\right)$$

$$(x_3 - \overline{x})(x_3 - \overline{x})^T = \left(\frac{1}{3}, -\frac{2}{3}\right)\left(\frac{1}{3}, -\frac{2}{3}\right)^T = \left(\frac{1}{9}, -\frac{2}{9}, -\frac{2}{9}, \frac{4}{9}\right)$$

Now, sum them up:

$$C = \frac{1}{3}\left(\left(\frac{1}{9}, \frac{1}{9}, \frac{1}{9}, \frac{1}{9}\right) + \left(\frac{4}{9}, -\frac{2}{9}, -\frac{2}{9}, \frac{1}{9}\right) + \left(\frac{1}{9}, -\frac{2}{9}, -\frac{2}{9}, \frac{4}{9}\right)\right)$$

$$C = \frac{1}{3}\left(\frac{6}{9}, -\frac{3}{9}, -\frac{3}{9}, \frac{6}{9}\right)$$

$$C = \frac{1}{3}\left(\frac{2}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{2}{3}\right)$$

$$C = \left(\frac{2}{9}, -\frac{1}{9}, -\frac{1}{9}, \frac{2}{9}\right)$$

Thus, the covariance matrix is:

$$C = \left(\frac{2}{9}, -\frac{1}{9}, -\frac{1}{9}, \frac{2}{9}\right)$$

iii) Find the principal components and variance along those principal components.

Solution:

iii) Principal Components and Variance

To find the principal components, we solve the eigenvalue problem for the covariance matrix C:

$$C = \left(\frac{2}{9}, -\frac{1}{9}, -\frac{1}{9}, \frac{2}{9}\right)$$

The eigenvalues are:

$$\lambda_1 = 0.333, \ \lambda_2 = 0.111$$

These eigenvalues represent the variance along the principal components. The corresponding eigenvectors (principal components) are:

$$v_1 = (-1, 1), \ v_2 = (1, 1)$$

Thus, the principal components are v_1 and v_2, and the variance along these components is 0.333 and 0.111, respectively.

Q11) The eigenvalues of a covariance matrix are $\lambda_1 = 7, \lambda_2 = 5, \lambda_3 = 1$

(a) What is the total variance in the data?

(b) What percentage of the total variance is explained by the first principal component?

The eigenvalues of the covariance matrix are given as:

$$\lambda_1 = 7, \lambda_2 = 5, \lambda_3 = 1$$

(a) Total Variance

The total variance in the data is the sum of all the eigenvalues:

$$\text{Total variance} = \lambda_1 + \lambda_2 + \lambda_3$$

$$\text{Total variance} = 7 + 5 + 1 = 13$$

(b) Percentage of Total Variance Explained by the First Principal Component

The percentage of the total variance explained by the first principal component is given by:

$$\text{Percentage} = \frac{\lambda_1}{\text{Total variance}} \times 100$$

Substituting the values:

$$\text{Percentage} = \frac{7}{13} \times 100 \approx 53.84\%$$

Thus, the first principal component explains approximately 53.84% of the total variance.

Q12) Consider a dataset of 2000 points all of which lie in $\mathbb{R}^{30}$. If all these points lie in a four-dimensional low subspace, then after applying the PCA algorithm which of the following statements is/are true?

(a) The covariance matrix corresponding to this dataset has 30 non-zero eigenvalues.
(b) The covariance matrix corresponding to this dataset has only four non-zero eigenvalues.
(c) The residues will become zero after four rounds.
(d) By utilizing four principal component vectors and their corresponding coefficients, we have the ability to reconstruct the complete dataset using representatives.

Solution:

We are given a dataset of 2000 points, all of which lie in a four-dimensional low subspace within $\mathbb{R}^{30}$. After applying Principal Component Analysis (PCA), we need to determine which statements are true.

1. The covariance matrix corresponding to this dataset has 30 non-zero eigenvalues.

- False: Since the dataset lies in a four-dimensional subspace, there will be only 4 non-zero eigenvalues. The remaining 26 eigenvalues will be zero.

2. The covariance matrix corresponding to this dataset has only four non-zero eigenvalues.

- True: Given that all points lie in a four-dimensional subspace, the covariance matrix will have rank 4. Thus, there are only 4 non-zero eigenvalues corresponding to the dimensions that explain the variance in the data.

3. The residues will become zero after four rounds.

- True: After projecting the data into the four principal components, the remaining dimensions (the residuals) will have zero variance because the dataset lies entirely within a four-dimensional subspace.

4. By utilizing four principal component vectors and their corresponding coefficients, we have the ability to reconstruct the complete dataset using representatives.

- True: Since the data lies in a four-dimensional subspace, all the information required to reconstruct the dataset can be captured by the four principal components and their corresponding coefficients.

Thus, statements 2, 3, and 4 are true, while statement 1 is false.